purchasing power parity. There are huge differences. In Panel A, the reserve demand for U.S. Treasuries is much higher than the market weights, which have been declining since 1985 apart from an upward tick caused by new, large issuances of debt after the financial crisis. In Panel B, market weights of Euro-area safe debt is well above reserve demand and higher than the output of the Euro area. Panel C shows the situation is reversed for Japan. Japan's market weight is about 30% at the end of the sample, but Japan's fraction of the world economy is less than 10% and shrinking.

Market weights are simply inappropriate for sovereign safe assets.

6.3. SAFE ASSET WEIGHTING SCHEMES

I compare the performance of some country-weighting schemes for government bonds for the set of sovereign issuers in the Citi Yieldbook database from 1990 to 2010.[47]

In Table 14.11, Panel A, I examine countries by various macro variables. Many macro variables predict sovereign returns: over the full sample, countries that are poorer, as measured by nominal GDP per capita, tend to have lower returns. There is a significant inflation risk premium; countries with higher inflation have higher sovereign bond returns to compensate investors for bearing higher inflation risk. Normalized measures of the amount of government debt predict returns, especially during and after the financial crisis.

In Panel B, I consider various weighting schemes from macro variables, along with market capitalization and GDP weights. Starting with market weights, I overweight (underweight) countries with favorable (unfavorable) values of the macro variable at the beginning of each year and track returns over the next year. The sign of the overweights is dependent on the sign of the predictive coefficients in Panel A. I report means and standard deviation of returns, along with raw Sharpe ratios (which do not subtract a risk-free rate, see chapter 2).

Panel B shows that market weights actually have done fairly well. We can do better by tilting toward countries that have faster growth, countries with high inflation, and countries with low debt/GDP ratios.[48] Since the financial crisis, GDP weights have done very well, with raw Sharpe ratios of 1.73 compared with 1.58 for market weights. This accounts for a large degree of the recent popularity

[47] I construct year-on-year returns in U.S. dollars with a constant duration of 5.0 taking only developed countries. The macro data are from the IMF. Regressions in Panel A use country fixed effects. In Panel B, I set the maximum deviation from the market weight at 5% and do not allow short positions. Thus a country with a market weight of 7% has weights between 2% and12%, and a country with a market weight of 2% is allowed to take weights between 0% and 7%. Rebalancing is done annually.

[48] The Debt/GDP measure is a nonlinear characteristic because very low measures are bad, but also very high measures are bad. Nonlinearity is much less pronounced in the developed country sample, which is used in Table 14.12.